MidiStruct MidiStruct

User Manual

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1. Installation

Step 1 — Download the file `MidiStruct_MidiStruct.lua`.

Step 2 — Copy it to your REAPER scripts folder:

- Windows: `%APPDATA%\REAPER\Scripts\`
- macOS: `~/Library/Application Support/REAPER/Scripts/`
- Linux: `~/.config/REAPER/Scripts/`

Step 3 — In REAPER, open `Actions > Show Action List`.

Step 4 — Click `New ReaScript`, browse to the `.lua` file, and confirm.

Step 5 — (Optional but recommended) Assign a keyboard shortcut to the action `Script: MidiStruct_MidiStruct.lua` for quick access.

No external dependencies. The script uses only the Lua API embedded in REAPER and standard ReaScript functions. Nothing to install.

Compatibility: REAPER 6.x and 7.x — Windows, macOS, Linux.

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2. First Launch — Quick Start

Open a blank REAPER project (or an existing one — tracks are added at the end, nothing is modified). Place the cursor at the timeline position where you want the track to start.

Run the script. Three dialogs will appear in sequence.

For a quick first test:

- Dialog 1: leave all values at default (`1,0,0,6,0`) → OK
- Dialog 2: type `2` (Pop vi-IV-I-V preset) → OK
- Wait 3 to 5 seconds

The script immediately creates 5 color-coded tracks, a timeline with named regions, and a text report in your project folder.

Press Play. You will hear the result with REAPER's default General MIDI instruments. Assign your own VSTi plugins to the 5 tracks to get the sound quality you want.

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3. Dialog 1 — Musical Styles

This first dialog contains 5 comma-separated fields.

```
Style (1=Pop 2=Tech 3=LoFi 4=RnB 5=House 6=Trap 7=Rock),Style 2 mix (0=None),Mix % (0-100),Complexity (1-10),Seed (0=Random)
```

Primary style (field 1 — value 1 to 7)

Sets the base style of the track. It affects BPM, swing, drum patterns, preferred scale, energy and tightness.

#	Name	BPM	Swing	Character
1	Pop	124	12%	Straight, catchy, major scale
2	Techno (Peak)	133	0%	Mechanical, tense, Phrygian scale
3	Lo-Fi Hip Hop	85	35%	Heavy swing, ride cymbal, pentatonic
4	R&B / Soul	92	28%	Groovy, Dorian scale, rich walking bass
5	Classic House	124	25%	4/4 kick, swung hi-hat, minor scale
6	Trap / Future	140	10%	Sparse bass, dark minor scale
7	Rock / Alt	120	0%	Open hi-hat throughout, high energy, pentatonic

Secondary style and Mix (fields 2 and 3)

Creates a hybrid style between two styles. Example: `1,4,40` creates a Pop/R&B mix with 60% Pop and 40% R&B. Numeric parameters (BPM, swing, energy) are linearly interpolated. Drum patterns and scale switch entirely to the dominant style.

Leave field 2 at `0` for a pure style with no blending.

Complexity (field 4 — value 1 to 10)

Controls the overall density of the melody and ornaments.

Value	Effect
1-3	Very sparse melody, few notes, minimalist
4-6	Balanced density — recommended starting value
7-9	Dense melody, many passing notes, frequent fills

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10	Maximum — may saturate some sections

Complexity also affects the frequency of drum ghost notes and chromatic passing notes.

Seed (field 5)

The seed is the number that initializes the random generator.

- 0 — generates a seed based on the system clock. Every run produces a different result.
- Any other number — identical result every run. Essential for retrieving a generation you liked.

The seed is shown in the text report and in the confirmation dialog.

Always note it down before re-running the script.

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4. Dialog 2 — Chord Presets

```
Preset number (0=Manual) 1-17
```

Type the desired preset number, or `0` to enter your own chords.

The 17 presets cover the most commonly used progressions in modern music production. See the full list in section 10.

Tip: even if you know your chords, try the presets first to understand how the engine responds before customizing.

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5. Dialog 3 — Manual Chords

This dialog only appears if you entered `0` in dialog 2.

```
Chords e.g.: Am:4 F:4 C:4 G:4
```

Basic syntax

```
NOTE QUALITY : DURATION
```

- Note: C, D, E, F, G, A, B with sharp (#) or flat (b)
- Quality: see table below
- :Duration: number of bars (optional, default = 4)

Chords are separated by spaces. The progression repeats automatically throughout the track.

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Available chord qualities

Syntax	Chord	Example
(none)	Major	C = C major
m or min	Minor	Am = A minor
maj7	Major 7	Cmaj7
m7	Minor 7	Am7
7	Dominant 7	G7
9	Dominant 9	D9
m9	Minor 9	Dm9
add9	Add9	Cadd9
sus4	Sus4	Gsus4
sus2	Sus2	Dsus2
dim	Diminished	Bdim
aug	Augmented	Eaug
5	Power chord	A5

Input examples

```
Am:4 F:4 C:4 G:4          → 4 chords of 4 bars eachDm7:2 G7:2 Cmaj7:4        → ii-V-I with unequal durationsA:4 D:4 E:4 E:4           → Rock blues in ACmaj7:2 Am7:2 Fmaj7:2 G7:2
```

Tip: short progressions (2 to 4 chords) generally give the most

coherent results. The engine can handle up to 12 chords (Blues 12 bars

preset) but the melody becomes harder to control.

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6. What the Script Creates in REAPER

5 color-coded MIDI tracks

Track	Color	Content
DRUMS	Red	Kick (36), Snare (38), Hi-hat (42/46), Ride (51/53), Crash (49)
BASS	Blue	Monophonic bass line, octaves 2-3
PAD-CHORDS	Purple	Voiced chords with voice leading, octaves 4-5, + CC#11
MELODY	Green	Lead melody, octaves 4-5, + CC#1 on long notes
COUNTER-MEL	Orange	Counter-melody, plays in MELODY silences

Color-coded timeline regions

Section	Color	Role
Intro	Dark blue	Drums only + minimal pad
Verse	Green	Full arrangement, medium density
PreChorus	Gold	Rising tension, hi-hat build
Chorus 1	Red	Original hook
Chorus 2	Red	Hook one octave higher
Chorus 3	Red	Hook + third harmonization
Bridge	Purple	Modulated +2 semitones, stripped arrangement
Outro	Teal	Progressive strip-down

Named MIDI items

Every item in the timeline is named `TRACK | SECTION`.

Example: `MELODY | Chorus 2`, `DRUMS | Bridge`, `BASS | Verse`.

This allows precise navigation and editing without opening items.

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Tempo automation

- Bridge: BPM - 2 (slight deceleration, breathing space)
- Chorus 3: BPM + 1 (slight acceleration, final energy)
- Outro: back to original BPM

Text report

Automatically saved to the project folder:

`MidiStruct_MidiStruct_rapport.txt`

Contains: date, style, inferred key, scale, BPM, swing, complexity, seed, number of bars, total duration, chord progression, hook description (rhythmic steps and intervallic contour), and full section-by-section structure.

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7. The 5 Tracks — Role and VSTi Setup

The script generates raw MIDI. The final sound quality depends entirely on the virtual instruments you assign to each track.

DRUMS

MIDI channel 10 (standard GM drum channel) for all drum notes.

All other channels contain nothing on this track.

Works natively with any GM drum plugin (ReaSamplOmatic, MT-Power Drumkit, EzDrummer, Superior Drummer, BFD, Steven Slate Drums, Addictive Drums…).

Notes used:

MIDI Note	Instrument
36	Kick drum
38	Snare
42	Closed hi-hat
46	Open hi-hat
49	Crash cymbal

51	Ride cymbal
53	Ride bell

If your drum plugin uses a mapping different from GM standard, use REAPER's MIDI routing or a note transposition item to remap.

BASS

Notes on MIDI channel 1, MIDI range 24-60 (octaves 2-3).

Works with any bass plugin: electric bass, synth bass, acoustic bass.

For a modern sound, a round-robin bass VSTi (Scarbee, Modo Bass, Trilian) will give the best results.

Note: the walking bass uses chromatic approach notes on the last step of each bar. These notes are short and soft — this is intentional. If your VSTi has a legato mode, enable it for smoother transitions.

PAD-CHORDS

Notes on MIDI channel 1, range 48-84 (octaves 4-5), with CC#11 Expression.

Important: for the exponential swell to work, your VSTi must respond to CC#11 (Expression). This is the case for most orchestral plugins (Spitfire, BBCSO, Kontakt libraries, EW Hollywood) and modern synthesizers.

If your synth does not respond to CC#11, you can either map CC#11 to the volume or expression parameter within the VSTi, or simply ignore the CC automation and work with static volume.

For jazz progressions (presets 9, 10, 11), an electric piano or Rhodes will respond naturally to expression. For EDM styles, a synth pad with a filter modulatable by CC#11 will produce organic movement.

MELODY

Notes on MIDI channel 1, range 48-84, with CC#1 Modulation on long notes.

CC#1 (Modulation) is mapped to vibrato in the vast majority of synth and orchestral VSTi plugins. If that is not the case for your instrument, check the LFO modulation configuration within the plugin.

For a vocal melody: a voice VSTi (Vocaloid, Synthesizer V, EAST WEST

Hollywood Vocals) with CC#1 mapped to vibrato will give very natural results.

For a more electronic sound: a lead synth with CC#1 mapped to vibrato

depth or filter cutoff.

COUNTER-MEL

Notes on MIDI channel 1, same range as MELODY but slightly lower.

The counter-melody plays exclusively in the silences of the lead melody.

It can receive the same VSTi as MELODY (for harmony) or a different

instrument (flute, violin, marimba…) to create a timbral dialogue.

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8. Working with the Seed

The seed is the key to the entire MidiStruct MidiStruct workflow.

Generating multiple versions

Run the script several times with seed = 0 (random). Each run produces

a different track with the same chord progression and the same style.

Note the seeds of the versions you like in the text report or in your

DAW via a project note.

Reproducing a generation

Enter the noted seed in the Seed field of dialog 1. The generation will

be rigorously identical, even weeks later, even on another computer.

Controlled variations

Keep the same seed but change one parameter (complexity, style 2, mix).

You get a variation of the same track with a slight musical difference.

This is useful for finding the exact balance between two styles.

Recommended workflow

1. Generate 10 to 20 versions with a random seed — note the seeds
2. Quickly listen to each version (30 seconds is enough)

3. Keep 2 to 3 promising versions
4. Reload the best one with its exact seed
5. Manually edit the elements that don't suit you
6. Use the rest as a production starting point

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9. Production Tips

After generation — what to do first

Assign VSTi before listening. REAPER's default General MIDI sound does not do justice to the generated MIDI content. Assign at least a decent drum kit to the DRUMS track and a synth to MELODY before evaluating the result.

Start by muting PAD and COUNTER-MEL. Listen first to DRUMS + BASS + MELODY. If this combination is already musically interesting, the track has potential. Then add PAD, then COUNTER-MEL.

Editing the melody

The MELODY track is designed to be edited. The script generates a musically coherent base — your manual interventions are the final 20%.

The most effective editing points:

- Extend resolution notes (step 14 of each A2/B phrase)
- Remove notes that are too close together and create unwanted clusters
- Decide whether to keep or delete chromatic passing notes
- Modify the climax at step 10 if the generated note doesn't suit you

Editing the bass

The walking bass generates chromatic approach notes on step 15. If an approach note sounds dissonant in your VSTi context, you can simply delete it — the bass remains coherent without it.

Regions as a mixing guide

The color-coded regions in the timeline reflect the track's dynamic exactly: Intro and Outro are at roughly 50% energy, Verse at 70%, Chorus at 100%. Use these regions to calibrate your volume automation faders and achieve a coherent overall dynamic curve.

CC Automation

The PAD track contains CC#11 curves. If you open the automation for this track in REAPER, you will see exponential curves on each chord. You can freely modify them — increase the swell_start_val for a more present pad, or reduce the peak for a more recessed pad.

The modulated Bridge

The Bridge is automatically transposed +2 semitones. If this modulation feels too abrupt in your context, you can edit the Bridge BASS track to bring it back to the original key — the Bridge will keep its borrowed chords but without the global modulation.

Hybrid style — recommendations

Combination	Recommended ratio	Result
Pop + R&B	60/40	Modern pop soul
Lo-Fi + Jazz	70/30	Jazzy lo-fi
House + Techno	50/50	Tech-house
Rock + Trap	60/40	Dark alternative
R&B + Pop	40/60	Emotional pop

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10. Reference — The 17 Presets

#	Name	Chords	Recommended style
1	Pop I-V-vi-IV (C)	C G Am F	Pop, House
2	Pop vi-IV-I-V (Am)	Am F C G	Pop, R&B
3	Pop I-IV-vi-V (C)	C F Am G	Pop

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4	50s I-vi-IV-V (C)	C Am F G	Pop, Rock
5	Minor i-VII-VI (Am)	Am G F G	R&B, Trap
6	Minor i-iv-VII (Am)	Am Dm G C	R&B, House
7	Andalusian i-VII-VI-V (Am)	Am G F E	Techno, Lo-Fi
8	Minor ii-V-i (Dm)	Dm Am E7 Am	Jazz, Soul
9	Jazz ii-V-I (C)	Dm7 G7 Cmaj7 Cmaj7	Jazz, Lo-Fi
10	Soul Am7 groove	Am7 D9 Gmaj7 Cmaj7	R&B, Jazz
11	Neo-Soul (Dm)	Dm9 Gmaj7 Cmaj7 Am7	R&B, Lo-Fi
12	Rock I-IV-V (A)	A D E E	Rock
13	Rock i-VI-III-VII (Am)	Am F C G	Rock, Pop
14	Rock Power i-VII-VI	Am G F G	Rock, Trap
15	Blues 12 bars (A)	A A A A D D A A E D A E	Rock, Lo-Fi
16	House vi-I-V-IV (Am)	Am C G F	House, Pop
17	Techno i-VI-III-VII	Am F C G	Techno, House

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11. Reference — Chord Syntax

General format

```
NOTE[QUALITY][:DURATION] NOTE[QUALITY][:DURATION] ...
```

Elements in brackets are optional.

Accepted notes

```
C  C#  Db  D  D#  Eb  E  F  F#  Gb  G  G#  Ab  A  A#  Bb  B
```

Accepted qualities

```
(none)   Major          Cm       Minor          Cmmin     Minor (alternative) Cminmaj7
Major 7          Cmaj7m7      Minor 7          Cm77       Dominant 7        C79       Dominant 9
C9m9      Minor 9           Cm9add9     Major add9       Cadd9sus4     Sus4             Csus4sus2
Sus2             Csus2dim      Diminished       Cdimaug      Augmented         Caug5       Power
chord        C5
```

Durations

Duration is expressed in whole bars. Default: 4 bars.

```
Am:4     A minor, 4 barsG7:2     G dominant 7, 2 barsCmaj7:8  C major 7, 8 barsF        F major, 4 bars (default)
```

Valid examples

```
Am:4 F:4 C:4 G:4Dm7:2 G7:2 Cmaj7:4 Am7:4A:4 D:4 E:4 E:4Eb:4 Bb:4 Gm:4 F:4Cm7:4 Fm7:4 Bb7:4 Ebmaj7:4
```

Important notes

Qualities are case-insensitive: `Am`, `am`, `AM` are equivalent.

Duration must be a positive integer. The separator between chords is a space.

The `|` character is ignored (useful for visual bar notation).

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12. Reference — GM Drum MIDI Notes

The DRUMS track uses channel 10 with the standard GM mapping.

MIDI Note	Instrument	Context in the script
36	Kick drum	All kick beat positions for the style
38	Snare drum	Beats 2 and 4, ghost notes, fills
42	Closed hi-hat	Regular steps outside open_steps
46	Open hi-hat	Steps defined by style (open_steps)
49	Crash cymbal	Start of each section
51	Ride cymbal	Lo-Fi and Lo-Fi Bridge (replaces hi-hat)
53	Ride bell	Lo-Fi, beats 1 and 3 only

Specific durations

Instrument	Duration (ticks)	Reason
Kick	110	Short, percussive attack
Snare	115	Slightly longer

Ghost snare	80	Very short, barely audible
Closed hi-hat	95	Standard
Open hi-hat	220	Longer for natural opening sound
Crash	800	Long for natural decay
Snare roll	55	Very short for roll density

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13. Frequently Asked Questions

Q: The script shows no dialog and closes immediately.

Make sure REAPER is running and a project is open. The script requires an active project to calculate the cursor position and the report file path.

Q: I hear drums but no melody or bass.

The MELODY and BASS tracks are on MIDI channel 1. If your VSTi is configured to listen only to a specific channel (other than 1 or "All"), notes will not play. Set the VSTi MIDI input to "All channels".

Q: Color-coded regions do not appear in my timeline.

Check that regions are enabled in REAPER: View > Show Regions.

On some versions, you also need to enable View > Regions Lane.

Q: CC#11 does nothing on my pad.

Your VSTi does not respond to CC#11 by default. Two solutions:

1. Within the VSTi, map CC#11 to the volume or expression parameter
2. In REAPER, use a MIDI filter (JS: MIDI Channel Filter) to route CC#11 to CC#7 (Volume) if your VSTi responds to CC#7

Q: The Bridge sounds dissonant compared to the rest.

The Bridge is automatically transposed +2 semitones. Some chord and key combinations can make this modulation feel abrupt. Solution: open the MIDI items on the Bridge BASS track, Ctrl+A to select all, then transpose -2 semitones (Shift+down arrow twice in the piano roll). The Bridge will keep its borrowed chords but in the original key.

Q: I want to reproduce a generation exactly but I didn't note the seed.

If you haven't closed REAPER since the generation, undo all actions (Ctrl+Z back to the blank state) then re-run the script — but with a new random seed you won't recover the original. Going forward: the seed is always visible in the text report MidiStruct_MidiStruct_rapport.txt in the project folder.

Q: Can I use multiple generations in the same project?

Yes. Each run adds 5 new tracks after the existing ones, with the corresponding regions and items. You can compare two versions side by side, or combine sections from different generations. Place the cursor at different positions so the tracks don't overlap.

Q: Generation takes a long time.

For short progressions (2-4 chords), generation takes 2 to 5 seconds. For the Blues 12 bars preset, it can take 8 to 12 seconds because the progression is longer and the CC#11 curve generates many points per chord. This is normal.

Q: Can I change the BPM after generation?

The initial BPM is set by a REAPER tempo marker at the start of the project. If you change the project BPM afterward, the MIDI will follow automatically (PPQ positions don't change, only the temporal interpretation does). The three automatic tempo markers (Bridge, Chorus 3, Outro) will remain proportionally consistent.

Q: How do I disable the Bridge modulation if I don't want it?

The modulation is not currently exposed as a parameter in the dialogs. To neutralize it: select all MIDI items on the Bridge, open the piano roll, Ctrl+A to select everything, then transpose -2 semitones.

Repeat on all 5 tracks (BASS, PAD, MELODY, COUNTER-MEL — DRUMS are not transposed).

MidiStruct MidiStruct — Algorithmic MIDI Generator for REAPER

ReaScript Lua — No external dependencies — Seed-reproducible

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